Congenital Analbuminemia (CAA): A Comprehensive Disease Characterization

Disease: Congenital Analbuminemia · **MONDO:** MONDO:0014449 · **OMIM:** #616000 · **Orphanet:** ORPHA:88918 · **Causal gene:** *ALB* (albumin), HGNC:399, located on chromosome 4q13.3

Summary

Congenital analbuminemia (CAA) is an ultra-rare, autosomal-recessive Mendelian disorder caused by biallelic loss-of-function (LoF) variants in the albumin gene, *ALB* (4q13.3; HGNC:399; reference transcript NM_000477). Affected individuals produce little or no serum albumin (typically <1 g/L, versus a normal ~35–50 g/L), yet—remarkably—the clinical phenotype in adults is usually mild. The paradox of "living without the most abundant plasma protein" is explained by robust physiological compensation: the liver increases synthesis of other secreted proteins (globulins, transferrin, fibrinogen, apolipoprotein A-I) to defend plasma colloid osmotic (oncotic) pressure, and albumin's many ligands (fatty acids, bilirubin, hormones, drugs) are redistributed onto lipoproteins and globulins. The dominant clinical features are therefore mild bilateral ankle oedema, fatigue, mild hypotension, and—most consistently—a secondary dyslipidemia (hypercholesterolemia, elevated beta-lipoproteins), which carries an increased long-term risk of atherosclerotic/coronary disease.

The genetic architecture is one of marked allelic heterogeneity: more than 20 distinct pathogenic *ALB* variants have been described, nearly all predicted to be null alleles (nonsense, frameshift, splice-site, and start-codon mutations acting through premature termination and nonsense-mediated decay). The single most common cause is the recurrent **"Kayseri" frameshift (c.228_229delAT, exon 3)**, which accounts for roughly half of all known cases and is found in geographically and ethnically distinct, often consanguineous populations—a founder effect. Only about 70 cases have been reported worldwide, making CAA one of the rarer Mendelian traits catalogued.

While adults are typically well, the **perinatal and childhood periods carry the highest morbidity**: recent pediatric cohorts reveal a multisystem picture of early-onset oedema, gastrointestinal morbidity (frequently misdiagnosed as protein-losing enteropathy), antenatal

complications, recurrent respiratory infections, dyslipidemia, and coagulation/immune abnormalities. Diagnosis rests on demonstrating near-absent serum albumin *without proteinuria*, followed by *ALB* gene sequencing. Management is supportive—lipid-lowering therapy, selective albumin infusion for acute needs (e.g., polyhydramnios in pregnancy), genetic counseling, and careful, free-drug-based dosing of highly protein-bound medications (the unbound drug fraction rises dramatically in the absence of albumin). The principal disease models are the spontaneous **Nagase analbuminemic rat (NAR)** and an engineered **Alb$^{-/-}$ mouse**, both of which recapitulate hyperlipidemia and compensatory hepatic protein synthesis while remaining broadly healthy.

Key Findings

1. CAA is an autosomal-recessive, allelically heterogeneous disorder caused by biallelic *ALB* loss-of-function variants (F001)

CAA maps unambiguously to the albumin gene *ALB* on chromosome 4q13.3. A landmark review by Minchiotti and colleagues sequenced *ALB* across the known case series and identified **21 distinct molecular defects**: seven nonsense mutations, seven splicing changes, five frameshift/deletions, one frameshift/insertion, and one start-codon mutation. Of the characterized probands, 19 were homozygous and one was compound heterozygous, consistent with a recessive trait that is typically unmasked by consanguinity or in restricted populations. The verified quote establishes the variant-class distribution:

> *"The twenty-one causative defects include seven nonsense mutations, seven changes affecting splicing, five frame-shift/deletions, one frame-shift/insertion and one mutation in the start codon."* — PMID: 23612153

Most mutations are private (unique to a single family), but one—the Kayseri allele—recurs across unrelated populations:

> *"Most mutations are unique, but one, named Kayseri, is responsible for about half of the known cases."* — PMID: 23612153

2. Variant spectrum: predominantly premature-termination (null) alleles acting via loss of function (F005)

The documented pathogenic *ALB* variants are overwhelmingly premature-termination alleles that are predicted to yield no functional protein, whether through nonsense-mediated mRNA decay or production of severely truncated, misrouted polypeptides. Representative variants include:

Variant (common name)	Location	Type	Note	PMID
c.228_229delAT (**Kayseri**)	Exon 3	Frameshift → premature stop	~50% of all cases; founder	12028999
c.1A>C p.Met1Leu (**Afula**)	Exon 1	Start-codon loss	Druze population, Galilee	23176518
c.527delC (**Erzurum**)	Exon 5	Frameshift, stop at codon 215	—	27346974
c.1098dupT p.Val367fsTer12	Exon 9	Frameshift/insertion	Consanguineous family	29981851
c.1309A>T	—	Nonsense	—	32881634
Arg114Ter (CGA>TGA)	Exon 4	Nonsense	Recurs (Swiss, American)	15996651
c.1652+1G>A	Intron 12	Splice-site → exon-12 skipping	Consanguineous	24627724
c.118+1G>A	Intron 1	Splice-site	Iraqi newborn	11781148
c.270+1G>T	Intron 3	Splice-site	German infant	26543026
Intron-12 + 3′UTR (compound het)	—	Splice + regulatory	Coronary thrombosis case	37184254

The frameshift mechanism is illustrated directly:

> *"causing a subsequent frame-shift with the generation of a premature stop codon, and an aberrant truncated putative protein product, p.Val367fsTer12"* — PMID: 29981851

Gene/protein identifiers: HGNC:399, NCBI Gene 213, UniProt P02768, reference transcript NM_000477. One case (Albumin Locust Valley) had a completely normal coding region, suggesting a remote regulatory-element defect (PMID: 16713595).

3. The adult phenotype is surprisingly mild and dominated by dyslipidemia and mild oedema (F002)

Despite near-total absence of albumin, most adults with CAA are clinically well. The consistent findings are a mild tendency to ankle oedema, asthenia, and moderate hypotension, with laboratory hyperlipidemia:

> *"The condition affects especially the lipid metabolism but apart from a possible increased risk for atherosclerotic complications, it is generally associated with mild clinical symptoms in adults."* — PMID: 23612153
>
> *"the large majority of analbuminemic subjects appear clinically well, the only consistent finding being a slight tendency to develop ankle edema. Asthenia and moderate hypotension have been recorded in several cases."* — PMID: 115079

Laboratory signatures include increased circulating cholesterol, beta-lipoproteins, and esterified fatty acids.

4. Pediatric CAA is a multisystem disorder with higher morbidity (F007, F002)

The perinatal/childhood window is the period of greatest risk. A 2026 genomic/phenotyping study of five children across two kindreds captured the multisystem nature of early disease:

> *"All patients presented with persistent hypoalbuminemia, early-onset edema, gastrointestinal morbidity (frequently leading to misdiagnosis as protein-losing enteropathy), antenatal complications, recurrent respiratory infections, and dyslipidemia"* — PMID: 42311911

The authors additionally documented hemostatic and immune abnormalities and advocated individualized monitoring for infectious and thrombotic complications. An earlier case corroborates the childhood pattern of oedema plus respiratory susceptibility:

> *"A 10-year-old white boy suffering from primary analbuminaemia presented with a 6-month history of ankle oedema. From infancy he had suffered from asthma and respiratory tract infections."* — PMID: 3983720

Suggested HPO terms: Hypoalbuminemia (HP:0003073), Edema (HP:0000969), Pedal edema (HP:0010741), Hypercholesterolemia (HP:0003124), Hyperlipidemia (HP:0003077), Recurrent respiratory infections (HP:0002205), Growth delay (HP:0001510), Abnormal bleeding (HP:0001928), Protein-losing enteropathy (HP:0100281), Fatigue (HP:0012378), Hypotension (HP:0002615).

5. The mild phenotype is explained by physiological compensation (F008)

Albumin normally provides ~75–80% of plasma colloid osmotic pressure and transports fatty acids, bilirubin, hormones, metals, and many drugs, while maintaining plasma redox balance:

> *"plays essential roles in regulating oncotic pressure, transporting endogenous and exogenous compounds, and maintaining redox balance"* — PMID: 42931509

In CAA, the liver compensates by up-regulating non-albumin secreted proteins, and ligands are redistributed—e.g., bilirubin accumulates in the high-density lipoprotein fraction. The compensation hypothesis is stated directly:

> *"the relative absence of severe illness in analbuminaemia reflects similar compensatory mechanisms afforded by the increase in non-albumin plasma protein levels and by other metabolic adjustments"* — PMID: 3983720

A historical series confirms that albumin's transport and nutritional roles are largely dispensable:

> *"there are now 14 reported cases of congenital analbuminemia in which serum albumin is absent or greatly reduced without clinical evidence of edema, decreased hormone levels or abnormal amino acid requirements"* — PMID: 686645

6. Cardiovascular/atherosclerotic complications are the main serious long-term risk (F006)

Chronic hyperlipidemia provides the mechanistic substrate for accelerated atherosclerosis. A young-adult case demonstrated the clinical reality:

> *"confirm that cardiovascular complications must be taken seriously in this pathology"* — PMID: 37184254 (35-year-old man with recurrent acute coronary syndrome/coronary thrombosis)
>
> *"apart from a possible increased risk for atherosclerotic complications"* — PMID: 23612153

7. Diagnosis and supportive treatment (F009)

Diagnosis rests on demonstrating near-absent serum albumin (documented values as low as 0.003 g/L; 81 mg/L; <8 g/L in neonates) with compensatory elevation of globulins and cholesterol. Serum protein electrophoresis/immunofixation shows an absent albumin band; confirmation is by *ALB* sequencing of the 14 exons (classically via SSCP/heteroduplex screening then direct sequencing). The key differential is *acquired* hypoalbuminemia:

> *"The clinical diagnosis may be challenging because of the absence of unambiguous symptoms and because hypoalbuminemia may have many causes different from a genetic lack of the protein."* — PMID: 26543026

CAA is distinguished from nephrotic syndrome, protein-losing enteropathy, hepatic failure, and malnutrition chiefly by the **absence of proteinuria** and the germline *ALB* defect. Treatment is supportive: statins for hyperlipidemia and albumin infusion reserved for acute needs—e.g., pregnancy polyhydramnios:

> *"Our patient suffered from polyhydramnios during her pregnancy which was successfully managed with albumin transfusions and had a normal vaginal delivery"* — PMID: 28154155

8. Drug pharmacokinetics are markedly altered: the free fraction of protein-bound drugs rises dramatically (F010)

Because albumin is the principal plasma drug carrier, its absence sharply increases the unbound (pharmacologically active) fraction of highly protein-bound drugs. In the Nagase analbuminemic rat, the free fraction of mycophenolic acid rose more than ten-fold:

> *"Free fractions of MPA in the control rats and NARs were 2.09 and 24.8%, respectively."* — PMID: 26643991
>
> *"Dose adjustments based on the total MPA may cause unnecessary overexposure to MPA in patients with hypoalbuminemia."* — PMID: 26643991

Loop-diuretic studies in NAR show that drug effect depends on residual globulin binding: furosemide binds albumin (and so loses efficacy), whereas azosemide, bumetanide, and torasemide bind globulins and retain activity (PMID: 24151919; PMID: 12516076). This is the single most clinically actionable consequence of CAA: **dose highly protein-bound drugs on free rather than total concentrations.**

9. Epidemiology: ultra-rare, recessive, consanguinity-enriched, founder effects (F004)

> *"only about seventy cases have been reported worldwide"* — PMID: 27346974
>
> *"CAA is rare, but its frequency seems to be significantly higher in restricted and minimally admixed populations."* — PMID: 23612153

Population frequency is well below 1 per 1,000,000 live births. Many reports arise from consanguineous families. The Kayseri allele has been found in Turkish, Amerindian, Arab, Slovak Romany, and Swedish individuals—consistent with either an ancient shared founder or a recurrent mutational hotspot.

10. Etiology is purely genetic; animal models and the AFP analogy (F003, F011)

CAA is a monogenic Mendelian trait with no environmental, toxic, infectious, lifestyle, or epigenetic cause, and no validated modifier genes or protective alleles. Phenotypic variability (severe neonatal vs. asymptomatic adult) is attributed to developmental stage and compensation rather than defined modifiers. The gene-family-redundancy concept is supported by the analogy to alpha-fetoprotein, a homologous albumin-family protein whose congenital deficiency is likewise benign:

> *"Because it is homologous to albumin, congenital deficiency of alpha-fetoprotein may be analogous to analbuminemia, a benign genetic trait."* — PMID: 1379776

The **Nagase analbuminemic rat (NAR)**, arising from a spontaneous *Alb* splice-site mutation in Sprague-Dawley rats, is the principal model. It recapitulates age-progressive hyperlipidemia and compensatory hepatic protein synthesis:

> *"All inbred strains of analbuminemic rats showed hyperlipidemia progressing with age"* — PMID: 3325299
>
> *"The liver, in an apparent compensatory response, increases synthesis of a group of secreted proteins defending plasma pi."* — PMID: 10432383 — NAR is described as "the only known condition in which transcription of both positive and negative acute phase proteins are increased simultaneously."

A TALEN-engineered knockout mouse mirrors the benign human trait:

> *"The resulting Alb(-/-) strains are analbuminemic but healthy."* — PMID: 25654695

Section-by-Section Characterization

1. Disease Information

CAA is an inborn error in which albumin, the most abundant plasma protein, is absent or severely reduced owing to biallelic *ALB* mutation. **Identifiers:** OMIM #616000; MONDO:0014449; Orphanet ORPHA:88918; MeSH "Analbuminemia." (Note: older literature cites OMIM #103600, an earlier/overlapping entry.) **Synonyms:** analbuminemia, congenital analbuminaemia, CAA, primary analbuminemia, hereditary analbuminemia. The information is aggregated at the disease level from case reports, small case series, molecular-genetic studies, and animal models—not from large EHR/individual-patient datasets (consistent with ultra-rarity).

2. Etiology

Primary cause: purely genetic—biallelic LoF variants in *ALB* (F001, F011). **Genetic risk factors:** homozygosity/compound heterozygosity for *ALB* null alleles; consanguinity and membership of restricted/founder populations markedly raise risk (F004). **Environmental risk factors:** none established. **Protective factors:** none genetically defined; a lipid-lowering diet may mitigate (but not cause) the secondary hyperlipidemia. **Gene–environment interactions:** none established; phenotype severity is modulated by developmental stage and compensation, not by exposures.

3. Phenotypes

Core phenotypes with suggested HPO terms and frequency qualifiers:

Phenotype	HPO	Type	Onset	Frequency
Hypoalbuminemia	HP:0003073	Lab abnormality	Congenital	Obligate (100%)
Hyperlipidemia / hypercholesterolemia	HP:0003077 / HP:0003124	Lab abnormality	Childhood–adult	Very frequent/ consistent
Pedal/ankle oedema	HP:0010741	Clinical sign	Any age	Frequent (mild)
Fatigue/asthenia	HP:0012378	Symptom	Adult	Occasional
Hypotension	HP:0002615	Clinical sign	Adult	Occasional
Recurrent respiratory infections	HP:0002205	Clinical sign	Childhood	Subset
Protein-losing enteropathy mimic / GI morbidity	HP:0100281	Clinical sign	Pediatric	Pediatric cohort
Growth delay	HP:0001510	Physical	Childhood	Subset
Abnormal coagulation/ bleeding	HP:0001928	Lab abnormality	Pediatric	Subset

Severity: generally mild in adults, variable/moderate-severe in neonates. **Progression:** stable/ chronic in adults; dyslipidemia is progressive. **QoL:** typically preserved in adults; pediatric/ perinatal morbidity may impair well-being.

4. Genetic/Molecular Information

Causal gene: *ALB* (OMIM gene 103600 → phenotype #616000; HGNC:399; NCBI Gene 213; UniProt P02768; NM_000477). **Variant classification:** nearly all reported variants are pathogenic null alleles (nonsense, frameshift, splice-site, start-codon). **Variant type/class distribution** (per 2013 review): 7 nonsense, 7 splicing, 5 frameshift/deletion, 1 frameshift/ insertion, 1 start-codon (F001/F005). **Allele frequency:** individually extremely rare; absent or near-absent in gnomAD in homozygous state. **Origin:** germline, recessive. **Functional consequence:** loss of function (no/absent protein). **Modifier genes:** none validated. **Epigenetic/chromosomal abnormalities:** none implicated (one case points to a possible remote regulatory-element defect, PMID: 16713595).

5. Environmental Information

No environmental, lifestyle, or infectious agent causes or triggers CAA (F011). Diet does not cause disease; dietary lipid management can modulate the secondary dyslipidemia. Not applicable: toxins, radiation, occupational exposure, infectious agents.

6. Mechanism / Pathophysiology — Causal Chain

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1. Biallelic LoF variant in ALB (4q13.3; nonsense/frameshift/splice/start-codon)
   → results in no functional albumin mRNA/protein (via NMD or truncation)
2. Loss of hepatocyte albumin secretion
   → leads to near-absent serum albumin (<1 g/L)
3. Absent albumin  →  branches:
   ├── (A) Loss of ~75–80% of plasma oncotic pressure
   │        → reduced colloid osmotic pressure (pi)
   │        → (i) mild interstitial fluid shift → bilateral ankle oedema
   │        → (ii) sensed by hepatocytes → COMPENSATION:
   │                ↑ transcription of non-albumin secreted proteins
   │                (globulins, transferrin, fibrinogen, apoA-I)
   │                → partial restoration of pi → mild, not severe, oedema
   │        → (iii) low pi stimulates hepatic lipogenesis (↑ACC, ↑FAS)
   │                + reduced lipoprotein catabolism
   │                → hypercholesterolemia / hyperlipidemia
   │                → (long term) atherosclerosis → coronary thrombosis/ACS
   ├── (B) Loss of ligand-transport function
   │        → ligands (fatty acids, bilirubin, hormones) redistributed
   │          onto lipoproteins/globulins → clinically tolerated
   │        → drugs: unbound (free) fraction rises sharply
   │          → altered pharmacokinetics/pharmacodynamics (dosing risk)
   └── (C) Loss of redox-buffering/antioxidant capacity (inferred)
            → reduced plasma antioxidant reserve (mechanistic, not fully
              demonstrated in CAA patients)
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Molecular pathways/processes: hepatic protein secretion (secretory pathway), de novo lipogenesis (acetyl-CoA carboxylase, fatty acid synthase; GO:0006633 fatty acid biosynthetic process), and regulation of oncotic/osmotic homeostasis. The hyperlipidemia mechanism was dissected in NAR vs. nephrotic rats: both show increased hepatic triglyceride secretion driven by reduced π, but proteinuria (absent in CAA/NAR) is required for the *catabolic* lipoprotein defect (PMID: 7723242; PMID: 1465061; PMID: 7611250). **Cell type:** hepatocyte (CL:0000182) is the central cell. **Protein dysfunction:** complete loss of function (no protein), not misfolding/aggregation/gain of function. **Immune involvement:** secondary (recurrent infections, immune abnormalities in pediatric cohort) rather than primary autoimmunity/immunodeficiency.

Suggested GO/CL terms: GO:0005576 (extracellular region), GO:0008289 (lipid binding), GO:0006633 (fatty acid biosynthesis), transport processes; CL:0000182 (hepatocyte).

7. Anatomical Structures Affected

Primary site of defect: liver (hepatocytes)—the organ of albumin synthesis (UBERON:0002107 liver; CL:0000182 hepatocyte). **Systemically affected compartment:** blood plasma (UBERON:0001969). **Secondary involvement:** cardiovascular system (atherosclerosis), lower-limb soft tissue (oedema), gastrointestinal tract and respiratory tract (pediatric morbidity). **Subcellular:** the hepatocyte secretory pathway—endoplasmic reticulum (GO:0005783) and Golgi (GO:0005794)—is where albumin is normally synthesized and secreted. **Lateralization:** oedema is bilateral/symmetric.

8. Temporal Development

Onset: congenital (albumin deficiency present from birth); clinical recognition ranges from neonatal to incidental adult diagnosis. **Onset pattern:** chronic/insidious. **Progression:** adult course is stable/chronic-lifelong; secondary hyperlipidemia is slowly progressive and age-dependent (demonstrated in NAR, PMID: 3325299). **Critical period:** the perinatal/childhood window is the time of highest vulnerability and intervention opportunity (F002, F007). **Course pattern:** progressive (lipids) on a stable baseline; lifelong. **Remission:** none spontaneous; the genetic defect is permanent.

9. Inheritance and Population

Inheritance: autosomal recessive. **Penetrance:** complete for the biochemical trait (hypoalbuminemia) in biallelic carriers; clinical expressivity variable. **Expressivity:** variable (severe neonatal to asymptomatic adult). **Anticipation:** not applicable (not a repeat-expansion disorder). **Founder effects:** yes—Kayseri (~50% of cases) and Afula (Druze). **Consanguinity:** frequently present; a major risk amplifier. **Carrier frequency:** unknown but very low; heterozygotes are clinically normal (mild hypoalbuminemia at most). **Epidemiology:** ~70 cases worldwide; prevalence well below 1/1,000,000. **Sex ratio:** no strong sex bias reported (autosomal). **Geographic/ethnic distribution:** enriched in restricted, minimally admixed, consanguineous populations (Turkish, Arab, Druze, Romany, Amerindian, and sporadic European cases).

10. Diagnostics

Laboratory: serum albumin <1 g/L (down to 0.003 g/L) with elevated globulins and cholesterol (LOINC: serum albumin 1751-7). **Serum protein electrophoresis/immunofixation:** absent albumin band. **Biomarkers:** near-absent albumin plus a dyslipidemia profile. **Confirmatory genetic testing:** single-gene *ALB* sequencing (all 14 exons + splice junctions + UTRs), historically preceded by SSCP/heteroduplex screening; WES/WGS or a hypoalbuminemia/liver gene panel are reasonable first-line in the genomic era. Carrier testing via targeted *ALB* sequencing enables counseling (PMID: 23176518). **Not applicable:** karyotype, FISH, CMA, mtDNA, repeat-expansion testing. **Differential diagnosis:** acquired hypoalbuminemia—nephrotic syndrome (proteinuria present), protein-losing enteropathy, hepatic failure, severe malnutrition, burns. CAA is distinguished by absent proteinuria + germline *ALB* defect (PMID: 26543026). A reported pitfall is concurrent acute glomerulonephritis masking the diagnosis (PMID: 18791744). **Screening:** not part of routine newborn screening; cascade/carrier testing is appropriate in affected families.

11. Outcome / Prognosis

Survival/life expectancy: generally normal in adults who survive the perinatal/childhood period; no established reduction in adult life expectancy. **Mortality:** increased perinatal/childhood morbidity and mortality is reported (PMID: 23612153). **Morbidity:** mild in adults (oedema, fatigue); multisystem in children. **Principal long-term threat:** accelerated atherosclerosis/coronary disease from chronic dyslipidemia (PMID: 37184254). **Prognostic factors:** age (perinatal most vulnerable), lipid control, and vigilance for thrombotic/infectious complications. **QoL:** typically preserved in adults.

12. Treatment

Management is **supportive**; no curative therapy exists. **Pharmacotherapy:** lipid-lowering agents (statins / HMG-CoA reductase inhibitors; NCIT C1456) for secondary hyperlipidemia. **Albumin replacement:** intravenous human albumin infusion (NCIT C29031 Albumin Human) reserved for acute indications (e.g., polyhydramnios in pregnancy, PMID: 28154155); not required as chronic therapy in most adults. **Pharmacogenomics/dosing:** dose highly protein-bound drugs on free (unbound) concentrations, because the free fraction rises dramatically (F010; PMID: 26643991); prefer diuretics that act via globulin binding (torasemide/azosemide/bumetanide) over furosemide. **Supportive care:** monitoring for thrombotic and infectious

complications, especially in children (PMID: 42311911). **Advanced therapeutics (gene/RNA/cell therapy):** none clinically available; theoretically plausible but not developed given the mild phenotype. **Experimental:** no registered CAA-specific trials identified.

13. Prevention

Primary prevention: not applicable (genetic/congenital); genetic counseling for consanguineous couples and known-carrier families reduces recurrence risk. **Secondary prevention:** early detection of affected neonates to anticipate perinatal morbidity; lipid screening and management to prevent atherosclerotic complications (tertiary prevention). **Genetic screening:** carrier testing and prenatal/preimplantation diagnosis feasible via targeted *ALB* sequencing in known families. **Immunization/public-health/environmental measures:** not applicable (non-infectious, non-environmental). **Counseling:** autosomal-recessive recurrence risk (25% per pregnancy for carrier couples); emphasize benign adult prognosis with careful drug dosing.

14. Other Species / Natural Disease

Taxonomy/orthologues: rat *Alb* (NCBI Gene 24186; *Rattus norvegicus*, NCBI Taxon 10116) and mouse *Alb* (NCBI Gene 11657; *Mus musculus*). **Natural disease:** spontaneous analbuminemia occurs in the Nagase analbuminemic rat (a splice-site *Alb* mutant). No established spontaneous companion-animal analbuminemia is catalogued in OMIA. **Comparative biology:** albumin belongs to the albumin/AFP/vitamin-D-binding-protein/afamin gene family on chromosome 4; loss of the homologous alpha-fetoprotein is likewise a benign trait (PMID: 1379776), supporting gene-family redundancy. **Zoonotic potential:** none (non-transmissible genetic trait).

15. Model Organisms

Nagase analbuminemic rat (NAR): spontaneous splice-site *Alb* mutant in Sprague-Dawley background; the Rat Genome Database (RGD) model of albumin deficiency. Recapitulates age-progressive hyperlipidemia (PMID: 3325299) and compensatory hepatic protein synthesis (PMID: 10432383); extensively used to study altered drug pharmacokinetics (PMID: 26643991, PMID: 24151919, PMID: 12516076, PMID: 31492695). **Alb$^{-/-}$ mouse:** TALEN-engineered knockout; "analbuminemic but healthy," used for human-albumin metabolism and albumin-based drug pharmacokinetics (PMID: 25654695). **Phenotype recapitulation:** both models faithfully reproduce the biochemical trait, hyperlipidemia, and compensatory protein

synthesis; they do not require the pediatric multisystem complications seen in some human infants (a limitation). **Applications:** oncotic-pressure physiology, hepatic lipogenesis, protein-binding pharmacokinetics, and albumin-drug carrier studies.

Mechanistic Model / Interpretation

CAA is best understood as a **compensated null-allele disorder**. A single lesion—biallelic loss of functional albumin—propagates into two clinically dominant downstream branches (oncotic and transport), each buffered by compensation:

Lost albumin function	Immediate consequence	Compensation	Residual clinical effect
Oncotic pressure (~75–80% of plasma π)	↓ colloid osmotic pressure	↑ hepatic synthesis of globulins/transferrin/ fibrinogen/apoA-I	Mild bilateral oedema
Oncotic → lipid homeostasis	↑ hepatic lipogenesis (↑ACC/FAS), ↓ lipoprotein catabolism	Partial, incomplete	Hyperlipidemia → atherosclerosis (main long-term risk)
Ligand transport (FA, bilirubin, hormones)	Carrier loss	Redistribution onto lipoproteins/globulins	Clinically tolerated
Drug transport	↑ free drug fraction	None effective	Altered PK/PD → dosing hazard
Redox buffering (inferred)	↓ antioxidant reserve	Uncertain	Not clearly demonstrated in CAA

The central insight is that **albumin's functions are individually dispensable because each is partially substitutable**, whereas no substitute exists for protein-binding of drugs—explaining why altered pharmacokinetics is the most reproducible and clinically actionable feature. The liver's capacity to simultaneously up-regulate both positive and negative acute-phase proteins in response to low oncotic pressure (a state otherwise seen only in analbuminemia; PMID: 10432383) is the physiological lynchpin of the mild phenotype.

Evidence Base

PMID	Title (abbrev.)	Contribution
23612153	*Congenital analbuminaemia: molecular defects and biochemical/ clinical aspects*	Core review: variant spectrum, Kayseri ~50%, mild adult phenotype, atherosclerotic risk, population enrichment
27346974	*A nucleotide deletion and frame-shift…*	~70 cases worldwide; Erzurum variant
12028999	*Molecular diagnosis of analbuminemia (Kayseri)*	Defines founder Kayseri c.228_229delAT
42311911	*Pediatric CAA multisystem phenotyping*	Multisystem pediatric phenotype; PLE mimicry; coagulation/immune signatures
3983720	*Analbuminaemia—South African patient*	Childhood oedema + infections; compensation hypothesis; bilirubin on HDL
115079	*Analbuminemia (quantitative albumin variant)*	Mild signs: ankle oedema, asthenia, hypotension; dyslipidemia
686645	*Hereditary disorders of albumin synthesis*	Albumin's transport/nutritional roles are dispensable
10432383	*Coordinate augmentation of liver secretory proteins*	Compensatory hepatic synthesis (NAR)
3325299	*Analbuminemic inbred rat strains*	Age-progressive hyperlipidemia in NAR
25654695	*Albumin-deficient mouse models*	Alb^{-}/$^{-}$ mouse "analbuminemic but healthy"
26643991	*MPA PK in NAR*	Free drug fraction 2.09%→24.8%; dosing guidance
37184254	*CAA with coronary thrombosis*	Cardiovascular risk in young adult; compound-het variants
28154155	*Pregnancy in CAA*	Albumin transfusion for polyhydramnios; drug-binding concerns
26543026	*Novel splicing mutation (German infant)*	Diagnostic challenge; differential diagnosis
1379776	*Congenital deficiency of alpha-fetoprotein*	Gene-family redundancy; benign-trait analogy

PMID	Title (abbrev.)	Contribution
7723242 / 1465061	*Lipid metabolism in NAR vs nephrotic rats*	Mechanism of hyperlipidemia; proteinuria required for catabolic defect
42931509	*Functional heterogeneity of HSA*	Defines albumin's normal functions (oncotic, transport, redox)
24151919 / 12516076	*Diuretic PK/PD in NAR*	Globulin- vs albumin-bound diuretic efficacy

Limitations and Knowledge Gaps

- **Small evidence base:** With only ~70 reported cases, all clinical generalizations rest on case reports and small series; no prospective natural-history cohort exists. Quantitative frequency estimates for individual phenotypes (e.g., what fraction develop coronary disease) are not well defined.
- **Perinatal mortality quantification:** Increased perinatal/childhood morbidity is repeatedly asserted but not quantified in population terms.
- **Modifier biology unknown:** The molecular basis of variable expressivity (severe neonate vs. asymptomatic adult) is not established; no modifier genes identified.
- **Redox/antioxidant branch is inferred:** Loss of albumin's redox-buffering role is mechanistically plausible but not directly demonstrated to produce clinical consequences in CAA patients.
- **One unexplained case:** Albumin Locust Valley has a normal coding region, implying an uncharacterized regulatory mechanism (PMID: 16713595).
- **Model gaps:** NAR and $Alb^{-/-}$ mouse capture biochemistry and pharmacokinetics but not the pediatric multisystem (GI/immune/coagulation) complications; carrier frequency and gnomAD-based population estimates were not directly computed in this investigation.

Proposed Follow-up Experiments / Actions

1. **Quantify population genetics:** Query gnomAD for *ALB* LoF allele frequencies to estimate carrier frequency and predicted birth prevalence, and to assess whether Kayseri represents a single founder haplotype vs. a recurrent mutational hotspot (haplotype analysis across ethnic groups).
2. **Build a natural-history registry:** Pool the ~70 published cases into a structured dataset (variant, age, phenotype, lipids, cardiovascular events, perinatal outcome) to derive phenotype frequencies and genotype–phenotype correlations.
3. **Cardiovascular risk study:** Systematically assess carotid intima-media thickness/coronary calcium and lipid trajectories in adult CAA patients to quantify atherosclerotic risk and test whether early statin therapy alters outcomes.
4. **Pharmacokinetic guidance:** Develop a free-drug-dosing reference for commonly used, highly protein-bound medications in CAA, extrapolating from NAR data and validating with patient free-concentration measurements.
5. **Molecular characterization of regulatory cases:** Pursue WGS/functional assays for coding-normal cases (e.g., Albumin Locust Valley) to identify regulatory-element lesions.
6. **Model refinement:** Phenotype the pediatric-relevant systems (coagulation, immunity, GI) in $Alb^{-}/^{-}$ mice/NAR to determine whether the childhood multisystem picture is modeled or is human-specific.